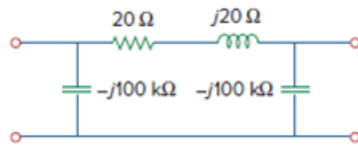


Problem

Find the transmission parameters for the circuit in Fig. 19.101.

Figure 19.101



Step-by-step solution

Step 1 of 10

Refer to Figure 19.101 in the text book.

Connect the voltage source V_1 as shown in Figure 1. Identify the currents and voltages in the circuit.

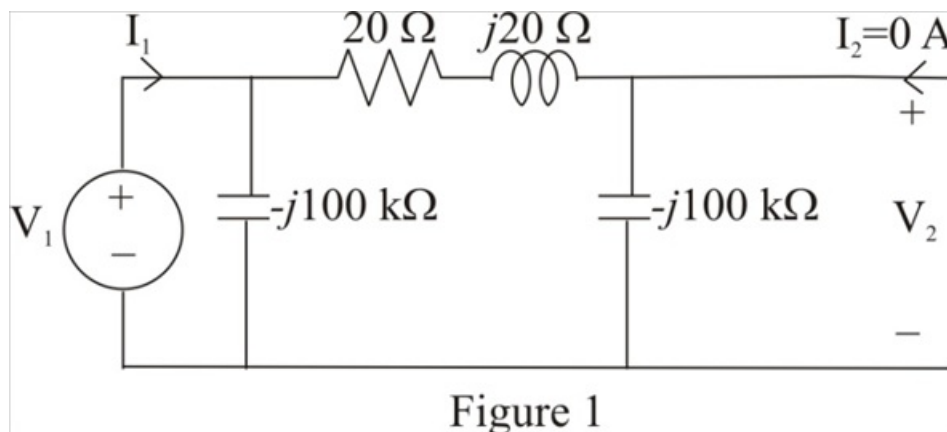


Figure 1

Step 2 of 10

Consider the expression for the voltage V_1 .

$$\begin{aligned} V_1 &= I_1 [(-j100 \text{ k}) \parallel (20 + j20 - j100 \text{ k})] \\ &= I_1 [(-j100 \text{ k}) \parallel (20 - j99980)] \\ &= I_1 \left[\frac{(-j100 \text{ k})(20 - j99980)}{(-j100 \text{ k}) + (20 - j99980)} \right] \\ &= I_1 \left[\frac{-99980000 \text{ k} - j2000 \text{ k}}{20 - j199980} \right] \end{aligned}$$

$$V_1 = I_1 (-40 - j499950)$$

Therefore, the expression for the voltage V_1 is $I_1 (-40 - j499950)$.

Step 3 of 10

Consider the expression for the voltage V_2 using current division rule.

$$\begin{aligned} V_2 &= I_1 \left(\frac{-j100 \text{ k}}{20 + j20 - j100 \text{ k} - j100 \text{ k}} \right) (-j100 \text{ k}) \\ &= I_1 \left(\frac{-10000000000}{20 - j199980} \right) \\ &= I_1 (-5 - j50005) \end{aligned}$$

Therefore, the expression for the voltage V_2 is $(-5 - j50005)I_1$.

Step 4 of 10

Determine the expression for the open circuit voltage ratio A .

$$A = \frac{V_1}{V_2}$$

Substitute $I_1(-40 - j499950)$ for V_1 and $I_1(-5 - j50005)$ for V_2 .

$$\begin{aligned} A &= \frac{I_1(-40 - j499950)}{I_1(-5 - j50005)} \\ &= 10 - j0 \\ &= 10 \end{aligned}$$

Therefore, the expression for the open circuit voltage ratio A is .

Step 5 of 10

Determine the value of the open circuit transfer admittance C .

$$C = \frac{I_1}{V_2} \text{ Substitute } I_1(-5 - j50005) \text{ for } V_2.$$

$$\begin{aligned} C &= \frac{I_1}{I_1(-5 - j50005)} \\ &= 0 + j0 \\ &= 0 \text{ S} \end{aligned}$$

Therefore, the value of the open circuit transfer admittance C is .

Step 6 of 10

Connect the voltage source V_1 and short the port 2 terminals as shown in Figure 2. Identify the currents and voltages in the circuit.

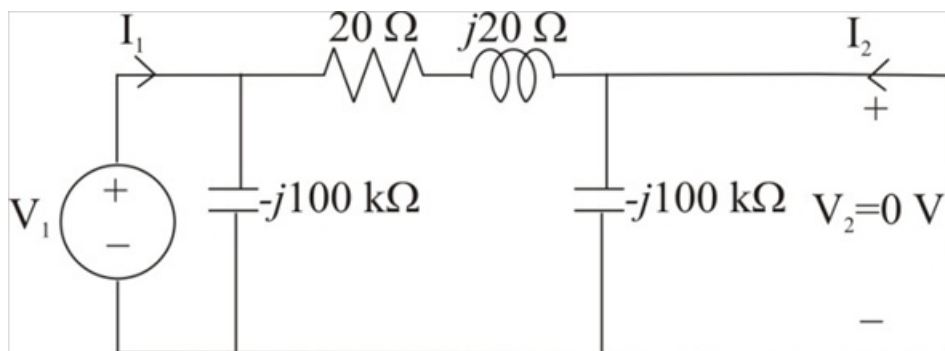


Figure 2

Step 7 of 10

Notice that the capacitor is shorted.

Consider the expression for the voltage V_1 .

$$\begin{aligned} V_1 &= I_1 [(-j100 \text{ k}) \parallel (20 + j20)] \\ &= I_1 [(-j100 \text{ k}) \parallel (20 - j20)] \\ &= I_1 \left[\frac{(-j100 \text{ k})(20 - j20)}{(-j100 \text{ k}) + (20 - j20)} \right] \\ &= I_1 \left[\frac{-2000000 - j2000000}{20 - j100020} \right] \end{aligned}$$

$$V_1 = I_1 (20 - j20)$$

Therefore, the expression for the voltage V_1 is $I_1 (20 - j20)$.

Step 8 of 10

Consider the expression for the current I_2 .

$$\begin{aligned} I_2 &= -I_1 \left(\frac{-j100 \text{ k}}{20 + j20 - j100 \text{ k}} \right) \\ &= I_1 \left(\frac{j100000}{20 - j99980} \right) \\ &= I_1 (-0.5 + j0) \\ &= I_1 (-0.5) \end{aligned}$$

Therefore, the expression for the current I_2 is $I_1 (-0.5)$.

Step 9 of 10

Determine the expression for the negative short circuit transfer impedance B .

$$B = -\frac{V_1}{I_2}$$

Substitute $I_1 (-0.5)$ for I_2 and $I_1 (20 - j20)$ for V_1 .

$$\begin{aligned} B &= -\frac{I_1 (20 - j20)}{I_1 (-0.5)} \\ &= (40 - j40) \Omega \end{aligned}$$

Therefore, the expression for the negative short circuit transfer impedance B is

$$\boxed{I_1 (40 - j40)}.$$

Step 10 of 10

Determine the value of the negative short circuit current gain **D**.

$$\mathbf{D} = -\frac{\mathbf{I}_1}{\mathbf{I}_2}$$

Substitute $\mathbf{I}_1(-0.5)$ for $\mathbf{I}_2$

$$\begin{aligned}\mathbf{D} &= -\frac{\mathbf{I}_1}{\mathbf{I}_1(-0.5)} \\ &= 2\end{aligned}$$

Therefore, the value of the negative short circuit current gain **D** is 2.